

Campus Course & Records Manager

Name: Arnav Sharma

Reg No: 24BAI10572

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Introduction

Universities and colleges handle large numbers of students, courses, and enrollments. Without a proper system, keeping records organized becomes difficult and inefficient. The *Campus Course & Records Manager (CCRM)* is a simple console-based Java application created to demonstrate how Object-Oriented Programming and core Java concepts can be used to manage academic information effectively.

The program allows users to add students, add courses, enroll students into courses, and display all records—making it a small but meaningful system for academic record handling.

Problem Statement

Managing academic records manually often leads to:

- Missing or misplaced student information
- Confusion between course codes and titles
- Difficulty tracking enrollments
- Time-consuming administrative work

This project aims to provide a simple and organized solution to manage these core academic elements:

students, courses, and enrollments.

Objectives

The primary objectives of this project are:

- To build a functional and menu-driven Java application
- To manage real-world entities: Students, Courses, and Enrollments
- To apply Object-Oriented Programming principles
- To implement data handling using ArrayList
- To practice input validation and exception handling

Methodology

The project is implemented as a **console-based program** where the user interacts with a menu repeatedly until they choose to exit.

Three main classes represent the system's core entities:

- **Student** – Contains ID, name, and email
- **Course** – Contains course code, title, and credits
- **Enrollment** – Links a student to a course

The **ArrayList** class is used to store students, courses, and enrollment objects dynamically. Input validation is implemented using `sc.hasNextInt()` and try-catch blocks to ensure the program does not crash on invalid input.

The workflow follows this sequence:

1. Show menu
2. Take user input
3. Perform the chosen operation
4. Repeat until user exits

Features

- 1 Add new Students
- 2 Add Courses
- 3 Enroll Students into Courses
- 4 View all Students
- 5 View all Courses
- 6 View all Enrollments
- 7 Input validation for safer user experience

Tools and Technologies Used

Programming Language: Java

JDK: Java 8+

IDE: Any Java IDE (VS Code, IntelliJ, Eclipse) or simple text editor

Core Concepts:

1. Classes & Objects
2. Encapsulation
3. ArrayList
4. Looping & Conditions
5. Exception Handling

System Design

The project uses three primary classes:

1. **Student** → stores student ID, name, and email
2. **Course** → stores course code, title, and credits
3. **Enrollment** → maps a student to a course

All operations are handled inside the main class **CCRM**, including user interaction and menu execution

Challenges Faced

During development, the following challenges were encountered:

- Ensuring valid numeric input during course credit entry
- Avoiding program crashes caused by invalid user input
- Searching for students and courses from lists efficiently

Learning Outcomes

Working on this project helped reinforce several important Java concepts:

- Designing classes to represent real-world entities
- Using ArrayList for dynamic data storage
- Implementing search operations within collections
- Handling exceptions and invalid user inputs
- Building structured, user-friendly console applications
- Improving logical thinking and debugging skills

Future Improvements

- File handling for permanent data storage
- Ability to update/delete existing records
- GPA calculations or course performance analytics
- GUI version using JavaFX or Swing
- Database integration (MySQL) for real-world scalability

Conclusion

The Campus Course & Records Manager (CCRM) is a simple but meaningful Java application that demonstrates how foundational programming concepts can solve everyday administrative problems. It successfully manages students, courses, and enrollments, and showcases clean program structure, data handling, and user interaction.

Demo

```
===== CCRM SYSTEM =====  
1. Add Student  
2. Add Course  
3. Enroll Student  
4. View Students  
5. View Courses  
6. View Enrollments  
7. Exit  
Enter choice: 1  
Student ID: 001  
Name: Arnav_Sharma  
Email: arnav.sharma.blr@gmail.com  
Student added!
```

```
===== CCRM SYSTEM =====  
1. Add Student  
2. Add Course  
3. Enroll Student  
4. View Students  
5. View Courses  
6. View Enrollments  
7. Exit  
Enter choice: 4  
---- Students ----  
001 - Arnav_Sharma (arnav.sharma.blr@gmail.com)
```